

BUZULUK, Russian FederationWMO: **289090**Lat: **52.8094N**Lon: **52.2278E**Elev: **80**StdP: **100.37**Time Zone: **5.00 (E05)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.3	-25.9	-32.4	0.2	-29.2	-28.8	0.3	-25.8	8.9	-6.5	8.1	-7.1	1.1	330	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	33.8	19.8	31.7	19.4	29.9	18.7	21.4	29.8	20.6	28.8	19.8	27.5	2.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	13.8	24.0	18.0	13.0	23.6	17.1	12.3	23.1	62.5	29.8	59.7	28.9	57.0	27.5	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.0	6.1	-32.4	36.2	3.5	2.0	-34.9	37.7	-36.9	38.8	-38.9	40.0	-41.4	41.4		
(5)			-32.5	23.3	3.5	1.7	-35.0	24.5	-37.0	25.5	-39.0	26.5	-41.5	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-11.5	-10.6	-4.1	7.2	15.1	19.5	21.3	20.2	13.7	5.8	-2.0	-8.8
	DBStd	13.23	7.27	7.12	5.55	5.90	4.81	4.56	3.60	4.39	4.73	4.79	6.06	7.38
	HDD10.0	2922	666	578	437	122	12	1	0	0	17	147	360	583
	HDD18.3	4963	924	811	696	334	123	39	13	31	153	388	609	841
	CDD10.0	1307	0	0	0	39	169	286	352	315	127	18	1	0
	CDD18.3	304	0	0	0	1	23	74	106	87	13	0	0	0
	CDH23.3	3474	0	0	0	24	313	856	1181	942	158	1	0	0
	CDH26.7	1361	0	0	0	2	86	350	477	402	44	0	0	0
Wind	WSAvg	2.6	2.8	2.9	2.9	3.1	2.7	2.4	2.3	2.2	2.2	2.5	2.7	2.7
Precipitation	PrecAvg	517	45	35	36	35	37	57	48	41	38	49	48	48
	PrecMax	698	95	56	62	90	117	129	95	86	114	88	119	96
	PrecMin	325	13	13	7	4	7	9	6	9	6	15	14	14
	PrecStd	99	16	11	15	21	26	35	26	21	24	21	25	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.7	2.1	11.8	26.3	31.1	35.5	35.7	36.2	31.0	21.5	11.6	3.1
		MCWB	1.0	0.8	7.4	14.3	17.5	20.0	20.2	19.8	17.4	12.6	10.0	2.1
	2%	DB	0.9	1.2	6.2	23.0	29.1	33.0	33.2	33.6	27.5	18.5	8.5	1.8
		MCWB	0.3	0.4	3.7	12.9	16.8	19.5	20.1	19.6	16.1	11.1	6.9	1.0
	5%	DB	0.0	0.4	3.9	20.1	27.1	30.7	31.3	31.0	24.7	15.7	6.6	1.1
		MCWB	-0.4	-0.2	2.0	11.6	16.2	18.9	19.8	19.1	15.2	10.0	5.3	0.5
	10%	DB	-1.4	-0.8	2.4	17.1	24.6	28.3	29.4	28.5	21.8	13.2	4.7	0.4
		MCWB	-2.0	-1.6	1.0	10.0	15.0	18.3	19.2	18.5	14.1	8.9	3.7	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.2	7.7	15.2	19.2	22.2	22.7	21.9	18.8	14.4	10.1	2.4
		MCDB	1.7	1.8	11.2	24.3	27.7	30.9	30.8	31.2	27.3	17.9	11.5	2.8
	2%	WB	0.4	0.6	3.9	13.6	17.9	20.8	21.7	20.7	17.1	12.3	7.2	1.2
		MCDB	0.8	1.1	6.1	21.6	26.5	29.9	29.6	29.8	24.9	16.9	8.1	1.5
	5%	WB	-0.4	-0.1	2.3	12.0	16.7	20.0	21.0	19.9	16.0	10.6	5.4	0.6
		MCDB	0.0	0.5	3.6	19.3	25.4	28.1	28.6	28.4	22.7	14.6	6.5	1.0
	10%	WB	-2.0	-1.5	1.2	10.4	15.6	19.1	20.2	19.1	14.8	9.4	3.8	-0.2
		MCDB	-1.4	-0.8	2.3	16.3	23.6	26.5	27.3	27.0	20.7	12.7	4.7	0.4
Mean Daily Temperature Range	5% DB	MDBR	6.7	7.9	8.0	10.2	12.3	12.2	12.2	12.3	11.6	8.3	5.2	6.0
		MCDBR	5.0	5.2	7.4	14.5	15.7	15.1	15.2	15.3	15.5	12.8	6.1	4.1
		MCWBR	5.2	5.0	5.4	6.8	6.6	5.5	5.2	5.2	7.0	7.3	5.1	4.0
	5% WB	MCDBR	4.9	5.2	6.8	13.5	14.6	13.1	12.9	13.6	13.8	11.1	5.5	4.0
		MCWBR	5.0	5.1	5.2	6.8	6.8	5.5	5.0	5.3	6.7	7.1	5.0	4.0
Clear-Sky Solar Irradiance	taub		0.301	0.325	0.373	0.429	0.406	0.401	0.408	0.427	0.382	0.351	0.328	0.301
	taud		2.062	2.033	1.983	2.083	2.270	2.340	2.316	2.248	2.338	2.334	2.239	2.083
	Ebn at Noon		640	749	789	797	845	852	837	795	789	732	620	562
	Edh at Noon		76	107	139	144	126	119	119	121	98	80	65	63
All-Sky Solar Radiation	RadAvg		1.02	2.14	3.54	4.48	5.68	6.17	6.06	5.01	3.36	1.71	0.76	0.61
	RadStd		0.14	0.17	0.40	0.48	0.45	0.66	0.52	0.42	0.46	0.30	0.13	0.15